Do US Firm Sizes Follow Zipf's Law?

In the Middle, Yes. At the Top, No

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Abstract

Axtell (2001, *Science*) reported that US firm sizes follow Zipf's law: the share of firms with at least s employees falls like $1/s$, from the smallest firms to the largest. We revisit the claim with the Census Bureau's public Business Dynamics Statistics for 1978–2023, plus Compustat for the largest employers. Between about 5 and 10,000 employees the law holds remarkably well in every year, with exponent between 0.94 and 1.00. It fails at both ends. There are too few firms with 1–4 employees, and, more strikingly, the largest firms are far too small. Zipf, extrapolated from the 1,210 US firms with 10,000+ employees in 2023, predicts a largest firm of about 12 million employees; the largest actual employer has about 2 million, worldwide. The tail exponent among the giants is about 1.3 today, against 1 for Zipf. By headcount it was 1.6–1.9 in earlier decades, so the largest employers have grown faster than the rest. Measuring size by sales or market value does not rescue the law at the top: on every measure the 50 largest US listed firms are too similar in size for Zipf.

1 The claim, and why it matters

Let S be the number of employees of a randomly drawn US firm. Zipf's law says

$$\Pr(S \geq s) \;\propto\; s^{-\alpha}, \qquad \alpha = 1.$$

Axtell (2001) fitted $\alpha \approx 1.06$ to 1997 Census data, from one employee to about a million, and found the same in other years. The law is a building block for theory. It is the benchmark outcome of random proportional growth (Gibrat's law with a lower barrier). It also governs whether firm-level shocks matter for the whole economy. With N firms and $\alpha = 1$, idiosyncratic shocks to large firms shrink only like $1/\log N$ in the aggregate, so a few giants can drive GDP fluctuations. With $\alpha > 1$ they wash out much faster (Gabaix, 2011, who measures firm size by sales). Whether the exponent is 1 or 1.3 is therefore not a detail.

2 Data

Census Business Dynamics Statistics (BDS, table `bds2023_fz`) gives, for every year 1978–2023, the number of US employer firms and their employment in ten size classes: 1–4, 5–9, 10–19, 20–99, 100–499, 500–999, 1,000–2,499, 2,500–4,999, 5,000–9,999 and 10,000+ employees. It covers all private nonfarm employer firms, 5.6 million in 2023. From it we compute $\Pr(S \geq s)$ at the nine lower class limits.

Compustat gives employment of US-incorporated listed firms. Employment there is worldwide, and private firms are missing, so we use it only for the shape of the very top, by fiscal year.

3 Results

1. The middle obeys Zipf. The log-log slope of $\Pr(S \geq s)$ between 5 and 10,000 employees lies between 0.94 and 1.00 in every year from 1978 to 2023: 0.99 in 1978, 1.00 in 1997 and 0.99 in 2023 (Figure 1, green). Over three orders of magnitude, a firm with ten times as many employees is ten times rarer. On this range Axtell's finding survives 26 more years of data.

2. The bottom does not. Only 60% of firms have 1–4 employees. If $\Pr(S \geq s) = 1/s$ held from $s = 1$, 80% would. Fitted from one employee, the exponent drops to about 0.66. Zipf describes firms above a few employees, not the whole population.

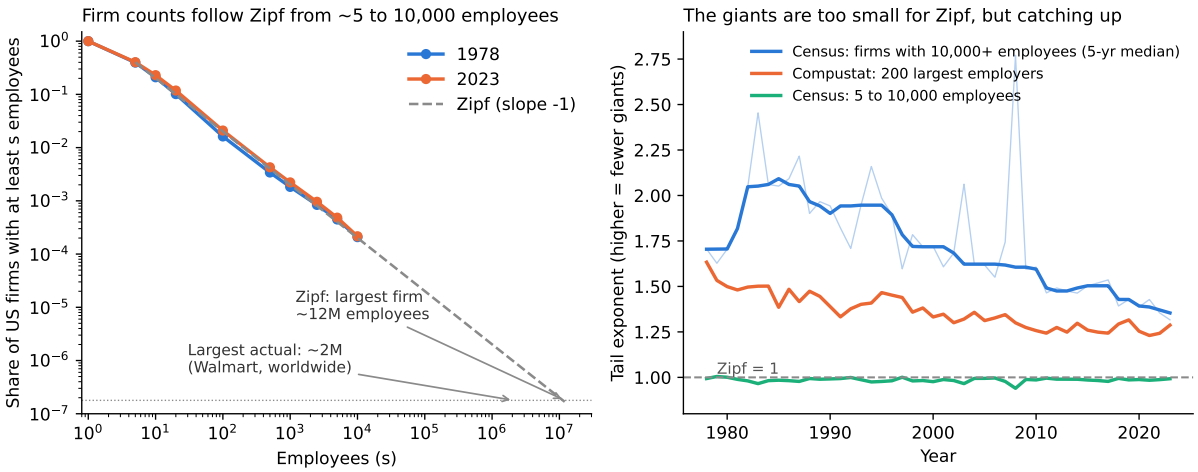


Figure 1: Left: share of US employer firms with at least s employees (BDS class limits), 1978 and 2023, and a Zipf line with slope -1 through the 5-employee point, extended to one firm. Right: tail exponents by year. Green: slope of the left-panel points from 5 to 10,000 employees. Blue: exponent implied by the mean employment of the 10,000+ class (Pareto truncated at 2.1 million; thin line annual, thick line 5-year rolling median). Orange: Hill estimator on the 200 largest Compustat employers. Zipf is 1.

Firms used	Size measure	1980–1999	2000–2009	2010–2023
Top 50	Employees	1.86	1.76	1.59
	Sales	1.66	1.78	1.66
	Market value	1.61	1.39	1.46
Top 200	Employees	1.43	1.32	1.26
	Sales	1.44	1.26	1.13
	Market value	1.26	1.10	1.17
Top 500	Employees	1.09	1.04	0.99
	Sales	1.11	1.02	1.00
	Market value	1.02	0.89	0.99

Table 1: Hill estimates of the upper-tail exponent of US-incorporated Compustat firms, averaged over fiscal years within each period. Each yearly estimate uses the largest k firms by the stated measure ($k = 50$, 200, 500). Employees are worldwide; sales are annual net sales; market value is fiscal-year-end price times shares outstanding. Zipf is 1; higher means fewer giants.

3. The top does not either. Two independent checks say the largest firms are much too small for Zipf.

- *The extrapolation check.* In 2023, 1,210 US firms had 10,000 or more employees. Under Zipf the r-th largest firm has about $10{,}000 \times 1{,}210/r$ employees, so the largest would have about 12 million. The largest employer in Compustat (Walmart) has 2.1 million, and that count is worldwide.
- *The mean check.* Firms in the 10,000+ class averaged 34,000 employees in 2023. A Zipf tail starting at 10,000 and capped at the largest firm would average about 50,000. Matching the observed mean requires a tail exponent of 1.32. The answer barely moves if the cap is 1.6 or 2.5 million instead of 2.1 million.

The 200 largest Compustat employers give the same answer, with a Hill estimate of 1.29 in 2023 (standard error about 0.09). The estimate depends on how deep one goes: in 2022 it is 1.57 using the top 50 and 0.97 using the top 500. The curve bends downward at the very top. It is not one straight line.

4. By headcount, the top has been moving toward Zipf. The exponent implied by the 10,000+ class was 1.7–1.9 (5-year medians) in the 1980s and 1990s and is 1.35 now. The Compustat Hill estimate fell from 1.63 in 1978 to 1.29 in 2023. The mean employment of the 10,000+ class rose from about 21,000 in the late 1980s to 34,000. The middle of the distribution did not change. The largest employers pulled away from it.

5. Does the measure matter? Headcount is Axtell's measure and the only one the Census tables offer, but it is not the only sensible one. A retailer and a software firm with the same sales can differ tenfold in

employees. Table 1 repeats the Compustat Hill estimate with sales and with equity market value, at three depths. Three things stand out. First, the 500 largest firms give an exponent close to 1 on every measure: well below the very top, listed firms look Zipf whichever way they are measured. Second, the 50 largest are too similar in size on every measure, with period averages between 1.4 and 1.9. In 2023 the largest firm by sales (Walmart) is 15 times the 100th largest; Zipf says 100. By employees the ratio is 26, by market value (Apple) 29. Third, the measures differ only in between. On the top 200, sales gave 1.13 in 2010–2023 against 1.26 for employees, and the decline over time is larger by sales. The headcount trend in point 4 is clearest among the top 50; by sales the top-50 exponent has not moved. With 50 firms a single-year Hill estimate has a standard error of about 0.2, so year-to-year changes at that depth are noise; the period means in Table 1 are more informative.

4 Caveats and related work

The BDS has only ten size classes, and the 10,000+ class mean pins the top exponent only if the tail is Pareto inside that class. That is why it is noisy year to year (for example 2008). Compustat counts employees worldwide, which makes listed giants look larger than their US workforce and so only strengthens the extrapolation check. It also omits private firms, which could thicken the very top. The BDS mean check covers all firms, listed or not, and gives the same answer. Kondo, Lewis and Stella (2023) use confidential firm-level Census data and conclude that firm sizes are "heavy tailed but not Zipf": a lognormal fits better than a Pareto with exponent 1, even far in the upper tail. A 2023 working paper re-examining the BDS reports a falling tail exponent through 2014. Our results agree with both and add a public-data update to 2023, with two checks anyone can redo in a few lines.

5 Why it matters

Zipf's law with exponent exactly 1 is the knife-edge case in which the largest firms matter most for aggregate outcomes. In US data that case holds for the body of the distribution but not for the giants, which are too few and too small. This is true by employees, the measure behind the original claim, and also by sales, the measure behind the granularity argument. Models calibrated to "Zipf all the way up" therefore overstate how big the largest firms should be relative to the next tier, and how much their idiosyncratic shocks should move the economy. The largest employers have grown faster than the rest since the 1980s, but the very top is still clearly thinner than Zipf on every measure we try.

A Code

Public data only (BDS table `bds2023_fz.csv` from `www2.census.gov`).

```
import csv, math
lo = dict(a=1,b=5,c=10,d=20,e=100,f=500,g=1000,h=2500,i=5000,j=10000)
D = {}
for r in csv.DictReader(open("bds2023_fz.csv")):
    D.setdefault(int(r["year"]), {})[r["fsize"][0]] = (float(r["firms"]), float(r["emp"]))
def alpha_from_mean(m, s=1e4, M=2.1e6, a0=1.0001, a1=5.0):
    f = lambda a: a/(a-1)*s*(1-(s/M)**(a-1))/(1-(s/M)**a)
    for _ in range(100):
        a = (a0+a1)/2; a0, a1 = (a, a1) if f(a) > m else (a0, a)
    return a
for y in (1978, 1997, 2023):
    d = D[y]; N = sum(v[0] for v in d.values())
    pts = [(lo[k], sum(d[j][0] for j in d if lo[j] >= lo[k])/N) for k in "bcdefghij"]
    x = [math.log(s) for s, _ in pts]; z = [math.log(p) for _, p in pts]
    mx, mz = sum(x)/9, sum(z)/9
    slope = -sum((a-mx)*(b-mz) for a, b in zip(x, z))/sum((a-mx)**2 for a in x)
    print(y, round(slope, 2), round(alpha_from_mean(d["j"][1]/d["j"][0]), 2),
          "Zipf largest firm:", round(1e4*d["j"][0]/1e6, 1), "million")
```

Output: 1978: 0.99, 1.71, 7.4 million; 1997: 1.00, 1.60, 10.0 million; 2023: 0.99, 1.32, 12.1 million.

References

[1] Axtell, R. L. (2001). Zipf Distribution of U.S. Firm Sizes. *Science* 293, 1818–1820.

[2] Gabaix, X. (2011). The Granular Origins of Aggregate Fluctuations. *Econometrica* 79(3), 733–772.

[3] Kondo, I. O., L. T. Lewis and A. Stella (2023). Heavy Tailed but not Zipf: Firm and Establishment Size in the United States. *Journal of Applied Econometrics* 38(5), 767–785.

[4] U.S. Census Bureau. Business Dynamics Statistics, 1978–2023 release.